Aiding the Communication of Aphasia Patients using Artificial Intelligence

Submitted September 2025, in partial fulfillment of
the conditions for the award of the degree **MSc Cyber-Physical Systems.**

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I hereby declare that this dissertation is all my own work, except as indicated in the text:

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Date 12 / 09 / 2025

Abstract

Aphasia is a language disorder caused by an injury to the brain, often caused by a stroke or trauma. This project explores the use of a pre-trained natural language model BERT (Bidirectional Encoder Representations from Transformers) and the adaptation necessary to analyse fragmented speech [1]. BERT uses the available context of a sentence to predict the contents of a masked word. This has been used to automate the repair of utterances where the speaker has struggled with word-finding and production. The AphasiaBank database [2] has been used to provide annotated transcriptions of language tasks, these labels have been used to locate error types, and have provided a suggested ground truth to be used to validate each target word prediction against a human's interpretation.

This study has focused on the contextual domains used to fine-tune the model and the size of neighbouring context necessary to produce an accurate prediction. This has explored the use of transcribed speech from participants with aphasia and healthy speakers, who have all conducted the same language tasks. The highest performing model has achieved an accuracy of 70% when trained on a combination of healthy and fragmented speech segments. This was observed when restricting utterances to a minimum length of 20 words. Since aphasic speech is typically disjointed, the augmentation of context from the speaker's neighbouring utterances can harm the model in some cases. Longer transcriptions may include frequent topic changes, areas of self-repair, and highly fragmented utterances from abandoned thoughts. Further investigation is needed to explore the impact of multi-modal context and how this differs from utilising additional text to support the predictions.

Contents

Chapter 1

Introduction

Aphasia is a common language disorder developed due to injury to the left side of the brain, often caused by a stroke, where the type of aphasia inflicted is dependent on the location and severity of the trauma. Aphasia has been shown to have a large impact on someone's quality of life, due to its deterioration of the production and understanding of language [3]. Intelligence and cognitive function are not affected [4]. This project focuses on those without cognitive conditions, such as dementia, who struggle to communicate their thoughts. This can be due to word-finding difficulties or, in the case of pre-existing dysarthria, muscle weaknesses that can worsen the production of speech [5].

1.1 Impact of Aphasia

Over the past 20 years, NHS England has observed a 28% increase among hospital patients admitted due to stroke events. This has been reported alongside a stark increase within the 50-59 age group, showing a shift in the at-risk demographic [6]. This increase in stroke cases has been theorised to be caused by common changes in lifestyle and cardiovascular health; which can cause blood restriction to the brain. In response, the NHS have launched the 'Act FAST' campaign to encourage awareness of the prevalent symptoms and the need for urgent medical response [6].

Approximately one-in-three patients experience 'post-stroke aphasia' [7]. In most cases, treatment is built on the assumption that the condition will naturally decrease with time, with language skills recovering a year after the stroke. However, full recovery is not always possible and long-term residual aphasia may continue to progress [8]. Treatment often targets the goal of "settling into a new way to be" and conducting exercises to re-strengthen the participant's conversational skills [9]. In this project, focus is placed on adapting the environment to ease communication during rehabilitation stages and assist issues that do not see complete improvement. This project aims to assist the understanding of an aphasia speaker's language by smoothing common structural issues and word finding errors within their utterances. This will relieve pressure and frustrations often reported with language disorders, and provide a sense of independence that can be lost on the diagnosis of a serious medical condition [9].

During investigations with speech-language therapists, researchers Northcott et al. explore the importance of an intertwined approach to providing psychosocial support and language therapy and the emotional impact this has on patients and their therapists.

This is due to the increased vulnerability that people with aphasia have to developing chronic depression [9]. Gillespie et al. [10] have explored the correlation between depression and stress caused by language difficulties. She brings attention to the controversy that is often involved in the diagnosis of post-stroke depression and how this harms the understanding of the condition. There are arguments that this depression is both under- and over-diagnosed; likely that those with aphasia are excluded from research due to the difficulties of adapting questionaries to accommodate their comprehension and communication needs [3].

1.2 Types of Aphasia

The various types of aphasia have different presentations due to the disorder affecting different language areas of the brain. Non-fluent aphasia produces halting, effortful speech [11]. A common symptom of this is agrammatism; this causes difficulty in structuring sentences, especially with respect to word ordering [4]. Although severity varies amongst patients, aphasia-produced utterances will often have simplified syntax and increased grammatical errors compared to healthy speech [12]. This results in 'telegraphic speech', composed mainly of nouns and verbs, leaving out smaller 'filler' words: such as articles, prepositions, and inflections [4]. This project focuses on the structural repair of these broken utterances to predict suitable replacements for missing words and ensure that the original intent is wholly conveyed.

Conversely, patients with fluent-aphasia are found to have noticeably smoother speech. However, this may not convey all intended meaning, often containing increased word substitution errors and nonsensical jargon [11]. Anomic aphasia creates difficulty with word-finding and naming tasks, with utterances being disjointed but likely following traditional syntax. Patients have been found to "work around" this issue, by describing the struggle-word or producing a similar word in its place. These patients are likely to benefit from simple therapeutic techniques, achieving successful retrieval when aided with a visual or spoken prompt [13]. This type of error can be repaired similarly by predicting the participant's original intended word.

Unlike other types of aphasia, Primary Progressive Aphasia (PPA) is a neurodegenerative disease caused by dementia, rather than the result of brain injury. This may present as fluent or non-fluent with symptoms that progressively worsen, often leaving the patient unable to produce verbal speech [14]. These patients often use other modes of communication, such as sign-language or assisted speech devices. The 'Aphasia Library' advise that these therapeutic tools are introduced as early as possible; to ensure a smooth learning curve before comprehension abilities further decline. In some cases, a person's own utterances can be used in computer-assisted speech production [14].

1.3 Project Scope

The developed model begins the repair of broken utterances, that have been produced by aphasia speakers, with an attempt to recover their intended meaning.

Inputs to model:
The model takes a transcribed utterance that has been manually annotated with error locations, as well as any prompts and supporting material: such as leading questions and visual aids.

Processing:
These utterances are cleaned to remove any syntax errors that have been transcribed, such as hesitation, stuttering, or repetition of words, to clearly isolate the intended speech of the patient. The labels are then used to identify substitution errors, and a Natural Language Processing model, BERT, generates a replacement suitable to the surrounding context.

The model focuses on the repair of the following common speech errors that may impede communication [15]:

1. filler words
2. repetition / preservation
3. jargon / neologism (nonsense-made up words)
4. semantic errors
5. circumlocution (indirect and round about words)

Desired output: The produced utterance is a direct translation of the original passage. Due to the ethical implications that come with modifying speech, it is important that the speaker is able to verify the produced speech and that all edits are highlighted to clearly indicate that there has been a change to what has been spoken. This is essential in the case where there is an erroneous divergence.

1.4 Literature Review

During recovery and treatment, it is recommended that those with aphasia visit a Speech-Language Pathologist who is able to identify the type of aphasia presented, and tailored rehabilitation techniques. For those with PPA, these therapy sessions focus on slowing the deterioration of language as symptoms progress. For other types of aphasia, therapy is motivated by the recovery of comprehension skills and the production of healthy speech [14]. The use of Artificial Intelligence to support language disorders is a popular area of research, often focusing on therapeutic or diagnostic tools. These assist with the rehabilitation of the patient through automated exercises that encourage the recovery of language skills, such as picture naming and word-recall.

1.4.1 Traditional Rehabilitation Methods

Physical resources can provide an alternative method of communication and remove speech barriers due to word production issues. These include both low and high technology communication aids, such as speech synthesis devices and pictograms - where the user can point to the pictures that represent the words that are causing difficulties [14]. These have been found to improve the lives of those with aphasia, but this is greatly limited by the willingness of the conversation partner, the technical support available, and the process time to produce messages [16].

Creating personalised therapeutic techniques has been found to contribute to a faster recovery, as the presentation of aphasia often varies within an individual over time and day-by-day [17]. Evaluative language tasks are used by a clinician to diagnose the severity of aphasia a patient has and the language areas that it is presented within, based on the types of errors presented [18]. Each exercise is designed to make use of different areas of the brain's language centre to identify the patients Aphasia Quotient - the type and severity of aphasia [19]. Tasks such as picture naming and narrative speech have been seen frequently within research. During these, the patient is presented with an image and is asked to recall the name of the subject and produce a relevant sentence or short story. This allows evaluation of the patient's ability to recall nouns, structure a sentence, and string together ideas [17].

1.4.2 Rehabilitation with Machine Learning

Structured exercises lend themselves towards machine learning classification tasks, to aid in the automation of diagnosis [12]; and binary evaluators to generate immediate feedback and indicate the most beneficial therapy approaches [18]. Aphasia rehabilitation may focus on the recovery of the patient's communication skills, or may place focus on external changes that can make activities easier. This may be through training conversational partners to adapt to the speaker's abilities, or translating works into aphasia-friendly text [20].

There is ongoing research on the use of machine learning technologies and artificial intelligence to make rehabilitation exercises accessible outside of a clinical environment. By introducing an element of autonomy to these tasks, the patient can receive unrestricted practice to improve their language skills. This should be used as additional support, rather than as a replacement to traditional techniques, as socialisation is highly beneficial for both language progress and mental wellbeing [21].

The research of Duc Le et al. [11] supports aphasia patients in their rehabilitation by providing personalised feedback of speech exercises. They have produced a web-based system to allow assessment of a person's word finding ability without the need for a clinician. Natural speech recordings are collected from the participant describing an image using predefined word prompts, and acoustic features are extracted from this. Keyword spotting is then used to identify production of the relevant words, and the patterns are compare against those from a correct and healthy response. Machine learning classifiers are used to estimate speech quality and these have been validated based on the scoring of human evaluators using four criteria "Clarity, Fluidity, Effort, and Prosody" [22].

1.4.3 AI Use in Healthcare

There is ongoing development into restoring language and communication skills through the production of computerised speech. Research teams such as Littlejohn et al. created a neural network mapping brain activity with the corresponding movement of muscles in the mouth to produce speech syllables [23]. This allows for the monitoring of brain signals to detect indication of an intentional utterance, which can be translated into a desired audible speech segment. This work is motivated by those who are unable to produce confident speech due to muscle weakness from conditions such as dysarthria; and creates a foundation for patients with damaged speech articulation. Additional research is needed into the application with brain injuries and how the received signals differ [24]. As part of this paper, Littlejohn et al. explore the challenges that arise when developing real-time bidirectional communication. They discuss the importance in minimising periods of silence while processing as this may disrupt the comfortable flow of a conversation. Therefore, continuous synthesis is necessary to emulate successful naturalistic speech [23].

This idea is supported by the work into conversational analysis by sociologists such as Harvey Sacks. A key topic during his Lectures on Conversation is the importance of sequencing in conversation and the idea of 'one-at-a-time' turn taking, which humans naturally fall into [25]. Large delays in speech production will lead to other participants attempting to complete these utterances due to the natural desire to progress the conversation. Filled silences must similarly be avoided, as these are naturally understood as an attempt to exit a conversation by breaking this progression [26].

The work by Binta et al. has explored the prediction of speech and word fill to complete utterances, alongside a study into the common issues observed within aphasia speech and how this differs between aphasia types [15]. This makes use of the AphasiaBank database of transcribed aphasic speech, which have been linguistically annotated by the interviewer. These are tokenised and passed into a pre-trained natural language model, BERT (Bidirectional Encoder Representations from Transformers), which predicts a masked word within an utterance using contextual information from the surrounding words. The authors conducted training at both sentence and paragraph level, performance was found to increase with size of input utterance [15]. When considering this alongside the findings of Littlejohn [23], it is necessary to decide whether to prioritise speed or context. For this stage of the project live speech production is unnecessary, therefore priority is placed on exploring the impact of available context.

Building from this research, the BERT model is used to generate a Natural Language Processing (NLP) model within out project. Due to its unified structure, BERT is able to adapt to a large number of NLP tasks with little modifications - such as sentence prediction and semantic analysis [1]. This project makes use of BERT as a Masked Language Model (MLM) for word prediction, to repair word errors within broken utterances. Since BERT is bidirectional, all words within the sentence are considered simultaneously to inform the prediction of the masked word - unlike traditional language models, which are limited to sequential processing [27]. This enables the model to be pre-trained on a much smaller amount of unlabelled data to achieve the same level of contextual embeddings. The BERT model undergoes an additional training stage to fine-tune its vocabulary and language patterns seen within the labelled domain. During testing, the model is self-supervised, making it well suited to process unlabelled live speech [28].

Chapter 2

Materials and Methodology

The majority of research is hindered by the lack of data available on aphasia speakers and the difficulty of its collection. Common speech recognition technologies (ASRs), such as commercial voice recognition software, are often inaccessible for users with language conditions. This is due to the disjointed nature of their utterances, often containing prolonged pauses and word errors which complicates addressing the system and selecting meaningful actions [29]. People with aphasia are likely to have a co-morbidity with dysarthria and apraxia dysarthria, leading to slurred speech or jargon which must be phonetically transcribed [30]. Transcription of aphasic speech is a key challenge in this area that limits the available datasets, as accurate transcriptions are necessary to produce a reliable translation conveying the speaker's intent. Work such as that of Provost et al. [31] has been conducted to begin the adaptation of ASRs to reliably interpret aphasia speech. Because of this, there is a need for the manual transcribing, which requires significant use of resources and introduces an element of interpretation which could inadvertently lead to a misrepresentation of the utterance [32].

2.1 AphasiaBank Data Source

The research organisation 'TalkBank' has created a shared multi-modal corpus collating speech data from participants with aphasia. This combats the common data availability challenges and facilitates research on the condition - funding exploration into the use of gestures as non-verbal communication and the presentation of aphasia within a speaker's non-native language [33].

Each dataset within the corpus contains transcriptions of interviews with speakers who have been medically diagnosed with all variations of the types and origins of aphasia, primarily resulting from a stroke. This allows observation into how presentation differs and explore interaction with pre-existing conditions, such as dyspraxia and dysarthria [34]. These interviews have been manually transcribed and are stored in multi-media form, such as video and audio clips. This project makes use of the written transcriptions to explore how spoken word can be adapted to smooth conversations and minimise miscommunication from both sides. Error indication markers are dependent on the codification used during transcription; the AphasiaBank corpus follows the CHAT standard (Codes for the Human Analysis of Transcripts) which is discussed in Section 2.2. The additional media available could be used in future project adaptations to extend this investigation.

Computer vision techniques and audio analysis would allow evaluation of the speaker's behaviour and emotions to strengthen inferences as to which utterances are likely to be erroneous.

2.1.1 Dataset Characteristics

For the initial evaluation stages of this project, datasets containing higher aphasia severities have been avoided. These contain challenging speech segments, with responses being highly disjointed and containing minimal utterances with little relevance to the question prompt. This structure becomes difficult to learn features of structure and reduces the phrases to extrapolate information from.

This project uses the Williamson Corpus [35] which contains data from 24 participants: 52% having fluent aphasia, 39% non-fluent aphasia, and the rest are undefined - with the majority of patients having Broca (26%) or Anomic (22%) aphasia. This database has been chosen due to its moderate level of word-substitution errors. This dataset has been partitioned with 70% of the transcripts used to train the model, and the remaining data is used for testing. The utterances have been contained within their transcripts to ensure the received context is individual to each speaker and conversation. From the Williamson dataset, 344 utterances are used from the 21 transcript gems - with 40% of the words randomly masked and predicted during training. The remaining 6 unseen transcripts have been used to test the model, with an average of 5 word errors per transcript, 29 of these utterances contain a mask token and 103 utterances are available to be used for context. Due to the small data set, the performance of the model is explored when treating each utterance in isolation and analysing the speaker's narrative response as a whole.

Control data from the Capilouto corpus [36] has been used to train the model on the typical language used within the contextual domain. This provides 72 transcripts of healthy speech, containing 1045 utterances in total. A comparison of the sequence lengths when split at an utterance and transcription level are given in Tables 2.1 and 2.2, these are used to vary the level of context passed into the model. This allows investigation into how the size of the context sequence impacts the accuracy of the predicted word repairs. The control speech contains a higher number of utterances and these are of a longer length, compared to the aphasia speakers.

Dataset	Utterance Count	Longest Size	Shortest Size	Average
Control Training	1045	30	1	6.7
Aphasia Training	344	22	1	4.6
Aphasia Testing	103	15	1	5.9

Table 2.1: Details on the input sequence lengths when processing transcripts at the utterance level.

Dataset	Speaker Count	Longest Size	Shortest Size	Average
Control Training	72	266	43	97.3
Aphasia Training	21	182	13	72.2
Aphasia Testing	6	221	59	99

Table 2.2: Details on the context lengths when processing utterances grouped by speaker and conversation topic. This provides maximum context.

2.1.2 Collection Protocol

The data collected within the AphasiaBank corpus has been collected against a standard protocol that is conducted following written script prompts. This allows for efficient collection of speech references that span a variety of areas of communication, while keeping the session conversational [37].

The participant is first asked to give a personal narrative of their stroke story and experience, this allows the collection of *free speech* samples and a reference to how the speaker engages with embedded questions. The remaining tasks use visual stimuli to encourage *picture description* and *narration* from sequential images and support recall of key events [37]. These tasks include the 'Refused Umbrella' and 'Broken Window' materials [38] which are shown in Figures 2.1 and 2.2; a corresponding script is given in Listing 2.1. The resulting responses are used as input to the project, alongside the supporting visuals to provide context and a method of evaluation. There is a greater difficulty to repair word-finding errors amongst free speech utterances due to the larger unknown contextual domain.

```
*INV: so — now I'm actually going to show you some pictures . and I'm
    gonna ask you to tell me a story &-um following them . and I need
    the story to have a beginning a middle and an end .
```

Transcript 2.1: The investigator's utterance to prompt a narrative picture description task paired with the 'Broken Window' image sequence (MMA002.cha).

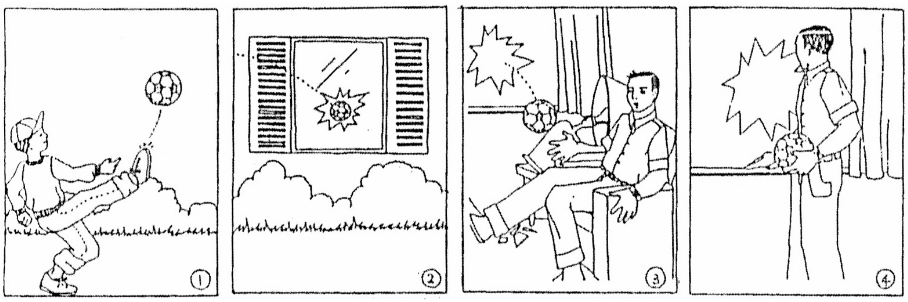


Figure 2.1: A six-panel sequence 'Broken Window' is used to prompt a descriptive narrative [38] [Lise Menn to AphasiaBank].

The last stage of the protocol assesses the key areas of speech affected by aphasia: repetition, noun and verb naming, and sentence comprehension. The patient's Aphasia Quotient is evaluated using the standardised *Western Aphasia Battery*, which provides an index of both the type and severity of aphasia [37].

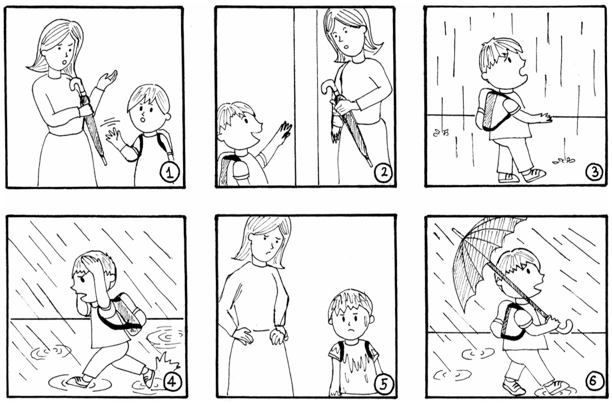


Figure 2.2: A six-panel sequence 'Refused Umbrella' is used within the protocol to prompt a descriptive narrative by the speaker [38] [commissioned by AphasiaBank].

2.2 CHAT Disfluency

AphasiaBank transcriptions follow the CHAT coding standard (Codes for the Human Analysis of Transcripts) from CHILDES [32]. Annotations and encodings are manually entered into the utterance as part of the transcription process. These are used during analysis of the individual's speech, for example: aggregation of word errors to evaluate progress and compare disfluency across aphasia types [32].

The need for manual transcription of aphasia speech introduces additional uncertainty to the data due to the necessary interpretation that may include an inaccurate portrayal of the utterances or wrongly classify errors. However, this human element of the transcription provides insight into how aphasia speech is understood and allows evaluation from both sides of the conversation.

2.2.1 Transcript Annotations

For the current project focus is placed on narrative speech and the misalignment between the participant's intended speech and what has been spoken. Many of the CHAT annotations are not relevant to this context and are removed to isolate the spoken utterances. This includes marking non-verbal actions with `\&=` as a prefix. For example: an event such as `\\&=laughs"`, a movement `\\&=turns:page"`, or a gesture `\\&=ges:unsure"` [32].

Other notations indicate the disjoint nature of the utterances, for example [32]:

- A "filled pause" or hesitance is marked using `\&-uh`.
- Unfilled pauses `(.)` `(..)` `(...)` mark silence and may encode the exact length.
- A period is used to denote end of an unmarked utterance - in aphasia speech several disjointed utterances are often combined to form a sentence.
- A trailing or incomplete utterance is marked with `+..` or `+.!` or `+!?`. Here the speaker may shift topic or forget what they were saying.
- Similarly, a self-Interruption is indicated `+//`.

- A phonological fragment with prefix `\&+` indicates where the speaker has abandoned a word or phrase.

These manual additions to the transcript can be used to highlight areas with low certainty and could benefit from repair to help communicate the intent of the speaker. These annotations reflect identified issues in the aphasia participant's production of speech and the understanding from their conversation partner.

2.2.2 Error Coding

Within written transcriptions, unclear segments or speech are highlighted following the standardised encoding style. These labels must be removed when cleaning the transcripts, but give an indication of the words that require translating.

The work of Jarvelin et al. [39] have monitored the response of dementia patients to word-naming tasks. From this, they have produced a neural network that models the language processing stages that have been witnessed. They have categorised word-production problems into four types:

"**Semantic errors** (*cat* → *dog*), **formal errors** (*cat* → *cap*), **neologistic (non-word) errors** (*cat* → *neet*) and **omission errors** (patient utters nothing or says "I don't know" etc.)" [39].

Within the CHAT encoding, suspected phonological errors are marked with `[* p]`, this is when the speaker has simplified speech sounds, such as replacing "boater for butter" [32]. Semantic errors are labelled with the `[* s]` prefix - this may be a mistaken word that is related to the target, such as swapping "mother for father"; or may be completely unrelated [32]. An example transcribed utterance has been given in Figure 2.2. These annotations can be used to flag low-confidence words that may need translating to produce an more comprehensible utterance. Errors from word and phrase repetition are indicated using a single forward-slash, with the implicated fragments enclosed in angle brackets. Example segments and other error encodings can be seen in Figure 2.3 taken from the The CHAT Transcription Format manual [32].

.,

```
*PAR: they cut the &+l lock off the door and tall [: call] [* p:w]
   the paramedics .
```

Transcript 2.2: An example transcript within the CHAT manual. This displays the standard noting of an erroneous word alongside its suggested repair. In this case a phonological error has been made

While the current prototype takes in the labelled transcripts of AphasiaBank, the final use case of the project will be performed on unlabelled live speech captured by an ASR device. Similarly, words and phrases that have been abandoned and revised by the speaker are indicated with a double forward-slash. This is a current limitation of the project that has not yet been accounted for and causes some discrepancies, for example, repairing of an erroneous word that has been abandoned or correctly self-repaired by the speaker. Detection of this anomaly could be automated by searching for adjacent equivalent word types, using Python's natural-language library NLTK.

Stuttering-like disfluencies (SLDs)	**Code**	**Example**	**Notes**
prolongation	:	s:paghetti	Place after prolonged segment
broken word	^	spa^ghetti	Pause within word
blocking	≠	≠butter	A block before word onset
repeated segment	↫	↫r-r-r↫rabbit like↫ike-ike↫	The ↫ brackets the repetition; hyphens mark iterations
lengthened repeated segment	↫ and :	↫r:-r:-r↫rabbit	Lengthening of the "r" segment
word repetition	[/]	dog [/] dog	Further distinctions are done inside FluCalc
Typical Disfluencies (TDs)	**Code**	**Example**	**Notes**
phrase repetition	<> [/]	<that is a> [/] that is a dog.	< > is used to mark repeated material
word revision	[//]	a dog [//] beast	Revision counts once
phrase revision	<> [//]	<what did you> [//] how can you see it ?	Revision counts once
phonological fragment	&+	&+sn dog	Changes from "snake" to "dog"
pause	(.) or (..) or (…)	(.)	Counts the number of short, medium, long pauses
pause duration	(2.4)	(2.4)	Adds up the time values, if marked
filled pause	&-	&-um &-you_know	Fillers with underscore count as one word

Figure 2.3: This table has been taken from The CHAT Manual, showing the coding style for a range of disfluency errors that are typically present in aphasia speech [32].

2.3 CLAN Software

The CLAN transcription software contains various commands to synthesise analysis from the available written data- for example, the 'COUNT' command returns the occurrences of speech fragments to compare across demographics [40]. This project makes use of the 'GEM' and 'FLO' commands to reformat the captured utterances and prepare the AphasiaBank for input into the model. This cleans the text of irrelevant codification elements, while preserving the structure and grammatical errors within the utterance. These commands are executed on a directory of transcript files, with each extracted Gem being fed into the FLO command. A Python file handler combines the outputs while separating each transcription with a delimiter to be processed in isolation.

2.3.1 GEM

The data collection protocol that has been followed by AphasiaBank contains multiple exercise stages. When documented, 'GEM' markers are used to signify the boundaries between each task - an example of this can be seen in Listing 2.4. A CLAN tool is used to extract the needed part of the session and isolate the speakers utterances corresponding to the prompt [40]. The command used for this is given in Listing 2.3, and an example output from the raw transcript data is shown in Listing 2.4.

```
gem +sUmbrella +n +d1 +t*PAR +r8 +u +f *.cha
```

Command 2.3: CLAN command to extract GEM

```
@G: Umbrella
*PAR: (..) the mother and the child . [+ gram]
*PAR: (...) ah &=finger:one . [+ exc]
*PAR: the mother tries to put the [/] the umbrella in the child's and
    [: hand] [* p:w] .
*PAR: &=head:no but the child fjufjuz@u [: refuses] [* n:k] it .
```

Transcript 2.4: An example raw transcription with Gem labelling to distinguish the segments of speech corresponding to the task. The full gem contains 15 utterances by the speaker (Williamson15a).

2.3.2 FLO TIER

The FLO tool is used to generate a cleaned version of the text based on the annotations provided by the human transcriber - errors follow the format `''spoken word [:suggested repair][* error type]''`, as can be seen in Listing 2.4. These are repaired by substituting the preceding text with material enclosed in the `'[:text]'` label - an example use of this command is given in Listing 2.5 and its output in Listing 2.6.

The repaired version of the transcript is used to train the model on the structure of aphasia speech, and to evaluate the model's generated words against the substitution suggested by the human transcriber. This allows for comparison against how a conversation partner infers the utterance, based upon their experience and full contextual knowledge available - such as word attempts. For some cases, word prediction may be used to fill silence and flesh out a disjointed utterance.

```
FLO +cr +t*  *.umbrella.gem.cex
```

Command 2.5: FLO command to return repaired transcription.

```
the mother and the child .
ah .
the mother tries to put the umbrella in the child's hand.
but the child refuses it .
```

Transcript 2.6: Example cleaned utterance taken from an aphasic transcript gem and the returned output using the FLO tool from CLAN (Williamson15a).

The FLO command given in Listing 2.7 uses a flag to toggle text replacement. This cleans the transcript of its annotations, while preserving the original words spoken by the speaker - including all errors, repetitions, and disjoint elements, while removing the irrelevant annotations. This allows extraction of the raw utterances to be used as input to the model, and mimics data that could be retrieved from an aphasia-friendly ASR. Example output from this command is shown in Listing 2.8.

```
FLO +cr +t* +r5 *.umbrella.gem.cex
```

Command 2.7: FLO command to return transcription cleaned of annotations and presevring raw speech using flag '+r5' [40].

```
the mother and the child .
ah .
the mother tries to put the the umbrella in the child's and.
but the child fjufjuz@u it .
```

Transcript 2.8: Example transcript cleaned from annotations and labelling but leaves errors as spoken (Williamson15a).

Chapter 3

Implementation

A masked-language model is used to predict suitable replacements for the word-finding errors, these are generated based on the surrounding context of the utterance, and the learned structure and vocabulary of sample aphasia speech segments. This work begins to repair broken utterances and assist in the communication of the speaker's intent. The transcripts have been taken from an AphasiaBank database and are passed into the pipeline, visualised in Figure 3.1, to prepare as input for the model. An illustration of this process performed on an example speech segment can be seen in Figure 3.2.

The raw utterances are extracted from the AphasiaBank corpus [2], and the following steps are executed upon them:

1. The FLO transcription tool, discussed in Section 2.3, is used to create a working copy of the transcript that is clean from annotation labelling. Basic errors such as repetition and filled silences are removed using Python's string pattern matching.

2. The encoded version of the utterance is searched for labels indicating word-errors, these are used to locate the error within the cleaned utterance.

3. The erroneous word is replaced by a mask token `'[MASK]'`; some words are decomposed into multiple masks due to the tokeniser's finite vocabulary. A MLM will be used to generate a suitable replacement word at this position.

4. When necessary, augmented utterances can be produced by stitching utterances with surrounding speech. This increases the context available to the predictor and can increase the accuracy of the generated word.

5. The utterance is tokenised using the BERT tokeniser.

6. A tensor structure is created to hold the produced token Ids of each word in the utterance and the human suggested error repair. These are used as ground truth values to evaluate the predictions against.

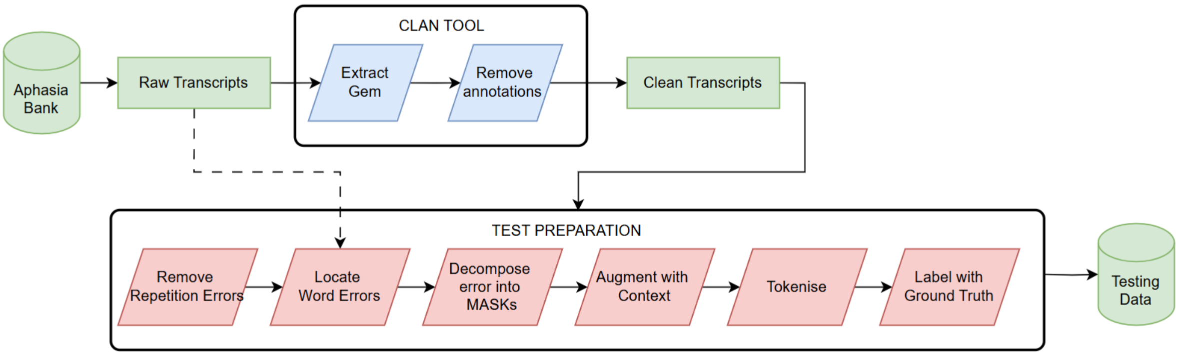


Figure 3.1: The pipeline followed to prepare testing data as input for the BERT model.

— Raw Utterance —

PAR: (.) she (..) puts the ʌnbɹɛlə˞ [:umbrella] [p:n] into is [: his] [* p:w] hands

— Cleaned output from FLO -—

she puts the ʌnbɹɛlə˞ into is hands .

— Located Errors —

Position of errors and their suggesed repairs: `[(3, 'umbrella'), (5, 'his')]`

— MASK Errors —

1. Split the repair into tokens: `['umbrella'] len=1`

   Replace with 1 mask token: `['she', 'puts', 'the', '[MASK]', 'into', 'is', 'hands', '.']`

2. Split the repair into tokens: `['his'] len=1`

   Replace with 1 mask token: `['she', 'puts', 'the', '[MASK]', 'into', '[MASK]', 'hands', '.']`

— Tokenise —

`[ 101, 2016, 8509, 1996, 103, 2046, 103, 2398, 1012, 102]`

with special tokens `[101] = [CLS]` and `[102] = [SEP]` to denote boundaries

— Label —

`[ -100, -100, -100, -100, 12977, -100, 2010, -100, -100, -100]`

— Running BERT —

`['she', 'puts', 'the', 'umbrella', 'into', 'his', 'hands', '.']`

Figure 3.2: Example output of the masking process, this demonstrates the tokenisation and labelling of the utterance that is necessary to create a supervised dataset.

3.1 Word Prediction for Repair

This project aims to assist the communication of people with aphasia by repairing common errors within their utterances, to reduce miscommunication and encourage socialisation. A Python module has been created to locate any word replacement errors in a transcription, and mask these to generate a suitable replacement with the BERT MLM.

3.1.1 Finding Position of Error

The FLO transcription tool is used to extract the spoken utterance from the encoded text; this is used as a basis for the model's input. The developed prototype takes advantage of the human-encoded text to identify the position of word errors in the utterance. This is implemented using the pattern matching of regular-expressions to search for the string marker `'spoken attempt [:suggested repair]'`. For the project's final usecase, the utterances received from the ASR would not have this labelling, and instead a confidence score may be attached to speech-fragments to indicate areas of error [41].

The Python module `diflib` [42] provides the `SequenceMatcher` class: this is used to identify matching subsequences between two strings. This is used to align the two versions of the utterance and locate the erroneous word within the cleaned text - this allows the suggested repair word to be extracted from the encoded utterance. An example alignment can be seen in Figure 3.4b. The class returns a `SequenceMatcher Object` containing tuples `(tag, i1, i2, j1, j2)` these variables are defined within Table 3.3 taken from the Python documentation [42]. These values are used to locate a substring with consecutive 'equal' and 'delete' blocks: this indicates an area where the utterances no longer match and the presence of an error repair. Pattern matching is used to verify the error type as a word replacement error, and the `(index, suggestion)` pair can be saved. This process has been illustrated by the example speech in Figure 3.4a.

Value	Meaning
`'replace'`	`a[i1:i2]` should be replaced by `b[j1:j2]`.
`'delete'`	`a[i1:i2]` should be deleted. Note that `j1 == j2` in this case.
`'insert'`	`b[j1:j2]` should be inserted at `a[i1:i1]`. Note that `i1 == i2` in this case.
`'equal'`	`a[i1:i2] == b[j1:j2]` (the sub-sequences are equal).

difflib — Helpers for computing deltas — Python 3.13.7 documentation [42]

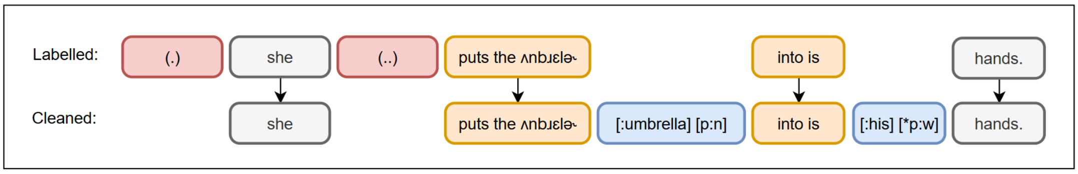
Figure 3.3: A table defining the possible values of the 'tag' variable returned from the diflib.SequenceMatcher class. These instruct how to "turn *string a* into *string b*" [42] and are used to align the located errors in the annotated and cleaned utterance.

Cleaned output from FLO : "she puts the ʌnbɹɛlə˞ into is hands ."

Labeled utterance : "(.) she (..) puts the ʌnbɹɛlə˞ [:umbrella] [p:n] into is [: his] [* p:w] hands ."

```
delete    a[0:4] --> b[0:0]    '(.) ' --> ''

equal     a[4:7] --> b[0:3]     'she' --> 'she'
delete    a[7:12] --> b[3:3]  ' (..)' --> ''

equal     a[12:31] --> b[3:22] ' puts the ʌnbɹɛlə˞' --> ' puts the ʌnbɹɛlə˞'
delete    a[31:50] --> b[22:22] '[:umbrella] [ p:n] ' --> ''

equal     a[50:58] --> b[22:30] 'into is ' --> 'into is '
delete    a[58:74] --> b[30:30] '[: his] [* p:w] ' --> ''

equal     a[74:80] --> b[30:36] 'hands ' --> 'hands '
equal     a[81:82] --> b[36:37]      '.' --> '.'
```

(a) Example output from the diflib.SequenceMatcher Python module.



(b) Alignment of the returned substrings. The blocks highlighted in orange and blue indicate a match and discrepancy pair that will contain a type of error. Other substrings with the 'delete' tag (shown in red) are caused by isolated annotations and can be ignored.

Figure 3.4: An example alignment of strings to indicate how to use the transcript annotations to find the position of the error word in the cleaned utterance.

3.1.2 Masking the Error

The cleaned versions of the utterances are transformed to create the model's input data. When tokenised, a specialised ID value is used to create the `'[MASK]'` token, this is inserted at the located indexes to hide the word-errors within the utterances and indicate the point of generation by the MLM. The BERT model uses the `WordPiece` tokeniser to encode words outside of its vocabulary based upon a decomposition of known 'subword' token IDs. Therefore, there is not always a direct word-to-token conversion, and multiple masks may be used as placeholders for the suggested repair [43]. For example, the word "unfurls" is masked into three tokens: `['un', '\#\#ful', '\#\#rs']`. For each mask token, the error index and suggested token ID are stored to be used as the ground-truth values during evaluation of BERT's predictions.

3.1.3 Labelling the Data

A structured tensor of labels takes the same shape as the tokenised utterance to evaluate each token against its ground truth. This holds blank values of -100 for each regular token (as this is ignored by the loss function), and the masked positions contain the ID value of the suggested repair. To account for the cases where the ground truth is decomposed into multiple tokens, each label must be offset by the number of masks inserted in the previous error - this ensures that the positions remain aligned.

In the final use case, the project will be used within a self-supervised environment to assist live communication. However, during the development of the prototype, the labelled data is necessary to evaluate performance [28].

3.1.4 Fine-tuning BERT

The word finding errors common within aphasia speech are repaired by predicting a suitable replacement using the BERT model. BERT is pre-trained on a general corpus by Google, and is trained on unlabelled data by randomly masking 40% of words in a passage and validating predictions against their known target values [43].

Additional model training is needed to fine-tune the generated repair predictions with a higher specificity to the relevant context. This uses a control database 'Capilouto' provided by the AphasiaBank corpus, using the FLO command to extract the spoken utterances from the recorded transcription. This contains responses from healthy speakers following the same protocol and picture stimuli. This allows the model to learn the domain-specific language used within the utterances [44]. Since aphasic speech is disjointed, further domain adaptation is necessary to specialise the model on the unique structure and language patterns of these utterances to produce more suitable word repairs. This additional training is executed on a partition of the aphasia dataset with the remaining data used for testing.

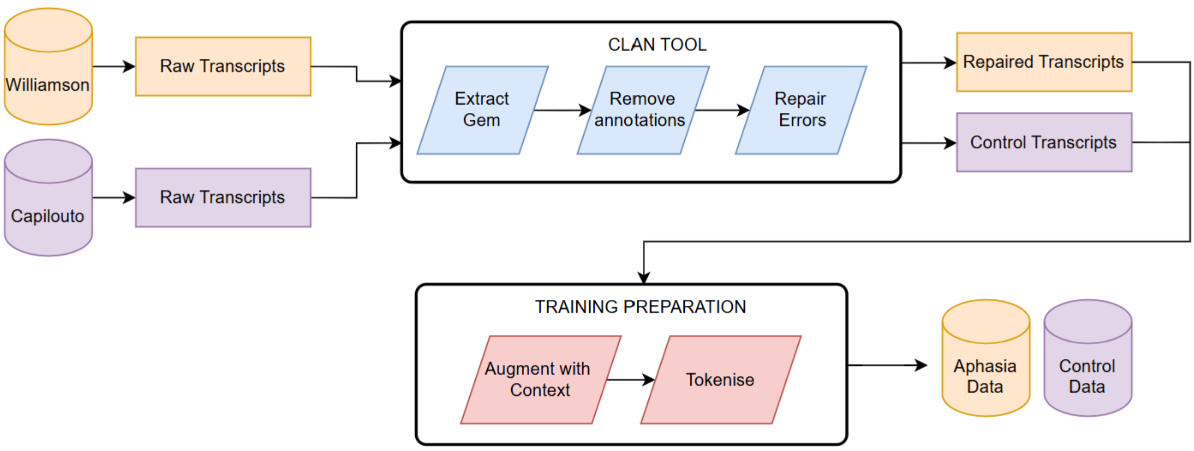


Figure 3.5: The pipeline followed to create the training data in a custom class.

During training stages, 15% of the input tokens are masked and predicted, using the AdamW optimiser, and model accuracy is calculated against the original data. A lower percentage of tokens are evaluated compared to the pre-training stage due to the smaller length of sequences used. This is repeated to predict random tokens within both the aphasia and controlled speech to replicate speech patterns from both [44]. There is an imbalance in the fine-tuning which may impact the model's ability to generalise. This is due to the size of the chosen datasets, which suffers from the natural higher availability of healthy speech compared to the challenges faced when collecting aphasia data.

A helper Python module has been created to congregate and format the training datasets, following the pipelined processes visualised in Figure 3.5. This reads from a directory containing extracted transcript passages and transforms these into a structured array. The utterances from different speakers are kept separate to ensure that irrelevant contextual information does not harm the loss function. Once tokenised, the values are transformed into custom data structures. The HuggingFace collator is used to randomly mask tokens and the data is loaded as a single input into the model [45].

Each mask replacement is generated in sequence to avoid the propagation of errors by ensuring predicted words are not considered during the generation of the next word [46]. Within the first tests, BERT is seen to reliably predict suitable replacements for small words, such as articles, prepositions, correct pronouns. The model struggles with words of higher specificity, particularly nouns and verbs; this is due to the lack of situation-specific data used within the pre-training sessions and the disconnect between the aphasic utterances and the structured speech that the model has been trained on.

3.2 Providing Context to Utterance

Training and testing performance has been evaluated by exploring the impact increasing context granularity has on the words generated. The AphasiaBank transcripts are split to create very short utterances based upon the natural breaks in speech caused by hesitation or delay; this reflects the speech production issues that are common within those with aphasia. The limited context available within these short utterances may make it impossible to predict a suitable repair word. To solve this, Binta et al. places a minimum word-length on their utterances to ensure that some context is always available to form a prediction [15]. When performing training at a paragraph level, Binta et al have found that larger dependencies between utterances provides richer context in learning language patterns and structure. However, this increases computational costs during the training stage due to the additional predictions during training and requires a longer process time, increasing the interruptions live conversations [23]. There is a risk of harming accuracy if utterances are augmented with unrelated topics or additional speakers [15].

With aphasic speech there are concerns with multiple word-finding errors occurring. For example, a speaker with anomic aphasia may produce the utterance `‘‘He takes the [MASK]"`. With the erroneous word masked, when given the visual prompt in Figure 2.2 in Section 2.1.2. As a valid repair, the sentence may be formed by replacing the masked word with any noun. A model with minimal context may generate any word with low confidence, such as ‘brush’ rather than the correct noun ‘umbrella’, which will shift the nature of the speech within any subsequent utterances. Therefore, the higher number of word errors present within the transcription, the less useful context will be available - this is especially true for neighbouring errors. There is also the concern of utterances being unrelated due to their disjoint nature: aphasia speakers are likely to change topic often, have periods of self-interruption abandoning an utterance, and may have circumlocution errors with run-on sentences. This would cause data to be augmented with irrelevant context and introduces ungeneralisable language patterns, which may harm the loss function and halt training progress.

Within the data structure, each transcript is separated to ensure context is added only within the boundaries of the particular participant's speech. A sliding-window is implemented to augment each masked utterance: the previous and trailing lines from the FLO transcript gem are appended until the minimum word length has been reached. At this stage, the repaired text is used for augmentation, rather than extracting a segment length from the spoken gem, to avoid masked tokens within the context. This may cause the propagation of errors and will be explored in a future testing stage.

3.3 Paraphrasing Circumlocution

This model produces a version of the speech with repaired semantic errors, focusing on the removal of repetition, the use of jargon, and word substitution errors. To repair any instances of syntax or grammatical errors; an additional model must be appended to the pipeline process. This can be achieved with a generalised extension of the BERT model: BART, which uses Bidirectional and Auto-Regressive Transformers to execute a wide range of language tasks [47]. A paraphrasing task can be used to remove elements of indirect speech and 'round about' errors where the speaker has backtracked and attempted self-repair. The example repaired transcription in 3.1 reflects a section of speech that has been severely impacted by aphasia. Although the participant has made few word substitution errors, it is clear that they have struggled with finding appropriate vocabulary and structure, leading to very short disjoint utterances:

```
this is her mother is going with the boy and go out .
he point show she
oh .
he didn't take it from her .
so he left home .
he's it's raining and he it it's raining .
look at it .
then he was home .
his mother said
he was he's
she was mad .
and she took the
then again he took a umbrella .
and on his head .
and walked away .
```

Transcript 3.1: An example transcription output from the BERT model where predictions replace each word error (Williamson01).

An example text generated from an instance of the BART model is given in Figure 3.2. This is using the summarization task from the BART pipeline, this has performed at too high abstraction and has taken away from the original speech. Although this is successful in explaining the intention of the participant, it has created a disconnection from the conversation that is taking place between the investigator and the speaker. Therefore, the produced text would not be useful within the project's main use-cases. For this prototype, focus is placed on communicating the meaning of the speech, rather than an exact replication, there is a need to preserve the natural elements of turn-taking and prompting conversational progress to create connection between its participants [25].

```
['summary\_text': '''It's raining and he it it's raining raining" is
   a children's book about a boy and his mother. In the book, the boy
    takes an umbrella from his mother and puts it on his head. When
   his mother gets mad at him, he takes the umbrella back.']
```

Transcript 3.2: An example output from the BART summarization task.

Instead of this, the model within Figure 3.3 has used the text generation task. This has produced a sentence that is instead a much lower abstraction of the spoken utterances. While this is closer to the model's intention, little changes have been made to the text to warrant the additional processing.

```
['generated\_text': ''This is her mother is going with the boy and go
    out. He didn't take it from her. So he left home. It's raining
   and he it's raining."]
```

Transcript 3.3: An example output from the BART general text generation task.

Further investigation is needed into the use of paraphrasing to repair circumlocution errors by fine-tuning this BART model through a similar approach to that taken has been with the BERT model. Evaluation of this must consider speech from a variety of patients with a range of aphasia quotients and severities. Since this will be performed on the output of the BERT model, it is of low importance to consider input text with repetition or word errors, though this should be explored in the case of erroneous repairs.

3.4 Multimodal Context

There has been research into taking additional contextual input from the visual resources, such as the material discussed in Section 2.1.2, but this has not been implemented to a level where a reliable accuracy score can be taken.

There are two key approaches that can be followed to implement this. The first technique extends the current vanilla BERT model by concatenating the image encodings into the input text tokens; this will inherently produce a lower accuracy than a dedicated multimodal model as BERT is not pre-trained on image tokens [48]. Similarly, an image-to-text model can be used to generate a caption summarising the content of the image, and this can be appended to the start of each input tensor [49]. The impact of introducing image context to the model has been tested using a small-scale prototype with the `'CLIP'` HuggingFace model, however, this has not reached a relevant accuracy.

The second approach uses a dedicated multimodal masked-language model as a replacement for BERT, such as `VisualBERT`, `FLAVA`, or `BLIP` [49]. An image-text-to-text model allows for a higher accuracy by taking both the image and repaired speech as separate inputs to produce the equivalent text [49]. Implementation of `VisualBERT` has been attempted using the `VisualBertForPreTraining` configuration provided by HuggingFace [48]. This provides both masked language modelling and sentence prediction based on the encoded image inputs. A basic image-encoding model, `CLIP`, was used to create these, however, this was unsuccessful as it is suggested to instead use a pre-trained CNN such as `ResNet`, to extract the key areas of importance from the image to create the visual embeddings [48].

Chapter 4

Evaluation

Model evaluation has been conducted throughout development of the project. This focuses on the adaptation of MLM prediction capabilities onto aphasic speech patterns, and explores the contextual levels required to produce accurate word generation. This is compared against baseline models and performance on healthy control language. During training, performance is based upon changes in the loss function and adaptive learning rates. The model's prediction accuracy is calculated considering the rate of produced exact matches with the human-suggested repair word and synonyms of this.

4.1 Evaluation Questions

The main focus of the conducted investigation explores how context granularity and input sequence length impacts the prediction of word-finding error repair. Domain context is provided through the augmentation of the training data; neighbouring utterances are stitched to each side of the speech fragment. A secondary investigation explores the contextual domains used for fine-tuning, to compare the benefits of training with a combination of healthy and aphasic speech, and the impact this has on learned language patterns.

The evaluation stage of this project takes advantage of the manual transcription of each utterance. Each word error is annotated with a suggested repair that the investigator believes is the speaker's intended word. Cases where a suggested repair has not been provided have been excluded from the dataset for the sake of evaluation. This evaluation metric cannot be exact as there is no way to know the exact intended speech of the participant. In most cases, this can be closely approximated by the human transcriber based upon conversational context and language experience. In addition to naming errors, necessary word repair may be caused by the use of incomprehensible language due to errors such as slurred speech, muscle weakness, or sound substitution. In the case where an audible attempt is available, phonetics and broken fragments can be extrapolated to deduce the target word.

Accurate transcriptions are necessary to create a reliable translation to correctly convey the speaker's intent. A faulty system is likely to introduce additional communication barriers and cause more frustration. A high error rate would introduce large ethical consequences as may distort utterances and incorrectly convey the speaker's intentions. There is a great risk of placing the responsibility of false words onto the speaker. For these reasons, it is essential to prioritise 'false negatives' and make no language substitutions if the confidence is below a minima threshold. The speaker should be given the opportunity to view and validate all system output before it is shared with their conversation partners.

4.2 Evaluation Function

Model performance is calculated considering whether the word has been replaced successfully and if this is a suitable replacement. An exact match is considered if the generated word is the same as the suggested repair made by the human transcriber. This match is preferred as implies that the model has been able to take the same information from the utterance as the conversation partner - who has access to a much larger level of personal context and experience. However, if the word generated has the same meaning as the intended word, it should also be accepted by the validator. This is as it is not possible to always confidently predict the exact intention, and instead the project focusses on conveying the original intent of the utterance.

Python's `Natural Language Toolkit (NLTK)` module is used to compare the produced word against the suggest repair. The `WordNet` lexical database, a "semantically oriented dictionary of English" [50], is used as a thesaurus of synonyms.

This is constructed as a hierarchy of words representing their relationships with similar concepts, a word is considered to be semantically similar if a path exists between them [50]. This can be discovered by generating `Synset Objects` of the words - with a value of 'None' being taken if the word is not present within WordNet's vocabulary - and calculating their Wu-Palmer similarity score. This computation is based upon the word's position in the hierarchy and their least common ancestor [50]. A linear loop is used to compare the definitions attached to each word from their WordNet Object to find the maximum similarity score, which is compared against a threshold.

The WordNet vocabulary does not include pronouns; for example, WordNet will incorrectly map pronouns onto similar nouns, such as `'I'` being confused for `'iodine.n.01'` and `'it'` defined as `'information\_technology.n.01'`. A separate evaluation is necessary to deal with these: pronouns are filtered by matching the appropriate 'piece-of-speech' tag, which labels each word based on their contribution to the surrounding sentence [51]. An additional challenge is introduced in analysing a semantic match against a pronoun. For example, a prediction of `'her'` may communicate the same intention despite a missing match against the suggested word `'mother'` - though this may be invalid if there is no relevant context to support this. This is a common feature of aphasia errors, as the participant may confuse pronouns or discuss subjects who have not been introduced.

Future development could explore methods that search for the prediction within the set of synonyms of the suggested repair. Analysis could be expanded to consider other words within the predicted set, these may be a more suitable fit to the sentence, even if the BERT model has produced a lower confidence score.

4.3 Comparative Baselines

Baseline accuracies have been computed for comparison and can be seen within the resultsin Section 5. The first uses the pre-trained BERT model available from the HuggingFace corpus that is intended for word prediction within healthy, standardised sentences. The second baseline used emulates potential general-public users, who are likely to use commercial AI tools, such as ChatGPT, rather than specialised LLMs. This is due to the ease of availability and capability to adapt towards the user's need case. Those with language disorders may struggle to navigate these without assistance as may have difficulties constructing effective prompts [52]. The responsibility is therefore likely to fall on the conversation partner or third party member who does not have aphasia or other language, or cognitive conditions. Many traditional communication tools, as discussed in Section 1, require training sessions to make effective use of these tools and overcome common learning curves.

```
Prompt: Please predict the masked word within this sentence, basing
    your prediction on only the surrounding sentence. Do not use
    previous passages to support this prediction.
```

Command 4.1: ChatGPT text prompt used to achieve produced basline - aiming to isolate tests across different speakers.

The chosen baseline model uses ChatGPT-5, this has adaptable computation and uses a "thinking" or "mini" variant depending on the processing required to respond to a task. When tested with maximum surrounding context, the model achieved an accuracy of 72% when allowed to use support from passages produced in previous prompts and other speakers. This decreased to 62.1% when asked to ignore the previous passages, the used prompt is given in Figure 4.1. While this does not make use of the LLM's full capabilities, it more accurately models the usecase: analysing a specific speaker within an isolated conversation. The words generated from these tests are given in Table 5.1 in Chapter 5, and can be found in the Supplementary Material.

When tested with minimal input sequence lengths, at an utterance level, the baseline model achieved an accuracy of 35.1%, and 29.7% when restricted to only using context from the given prompt. Although it is not clear whether the additional context is always avoided, as sometimes the chatbot has ignored this part of the prompt. This is as generative chatbot models are often most successful when able to use data from all available chats to personalise responses. Further tests should be done to fully take advantage of this; however, it is important to recreate performance on a single speaker's transcription to emulate use within a natural conversation and their own aphasia quotient.

The ChatGPT model has not restricted itself to one-to-one prediction with each mask token within the sentence. For example, in response to the utterance `‘‘[MASK] [MASK] [MASK] running home in the rain"` it has produced the sentence `‘‘[he][is] running home in the rain"`. This is a synonymic match to the manual repair `‘‘[he][’][s] running home in the rain"`, and has been produced by treating the three mask tokens as a single repair task - rather than computing each sequentially. To reproduce this in the developed prototype a more advanced version of the BERT model is needed, such as SpanBERT, to predict sequential substitutions with higher accuracy.

4.4 Training Performance

Three epochs have been performed on data batches of 16 and 32. For iterations with minimal sequence lengths, the bare utterance data sequences are very short, often only a few words, and give minimal assistance in the learning of general language patterns. A low number of tokens (40%) have been masked and predicted for training, resulting in very sparse and noisy gradients, which has caused training instability. There has been large over-fitting in the first epoch and collapse of the loss function by the third epoch, with values of `[0.034, 1.83, NAN]`. While the `ADAMW optimiser` is well-suited for these sparse gradients, further improvement has been found through a scheduled learning rate multiplier to adapt to these changing gradients.

In other iterations of the model, the entirety of the speaker's transcript gem, or an exert reaching the maximum sequence length, can be used to train the model with sequences of maximum available context. This results in a loss function that is expected for a MLM, decaying at a suitable rate of `[0.781, 0.612, 0.575]` across three epochs.

Chapter 5

Discussion

The primary investigation has focused on the adaptation of word prediction models to be used on disjoint aphasiac utterances and the impact available contextual granularity has on the accuracy of the generated repairs. The predicted words are used to replace the spoken word errors and aid in the communication of intentions. Context is introduced to the input utterances by placing a sliding window on the transcription to attach neighbouring lines to either sides of the utterance to reach a minimum sequence length. This data augmentation is performed on both the training and testing sets.

A secondary investigation observes the impact of the nature of the data used during the training stage. Models have either been fine-tuned with a control dataset containing healthy responses, or a repaired partition of the aphasia dataset. The use of healthy speech allows the model to identify context-specific vocabulary likely to occur within the tests, while the use of inflicted speech provides learning into language patterns that are specific to and common within aphasia speech. Evaluation of the system considers combinations of the available contextual data to investigate how fine-tuning the BERT prediction model impacts the generated accuracies. These scores are compared with baseline performances taken from OpenAi's ChatGPT v5.0 responses and the pipeline BERT model provided by HuggingFace and pre-trained on the Google corpus [28].

The current testing stage has focused on a single transcription Gem of the collection protocol, where the speaker provides a narration based on the 'Refused Umbrella' visual prompt given in Section 2.1.2 and Figure 5.1. This contains six sequential illustrations of a mother and child preparing for a walk in the rain.

Testing of each model version has been performed with the remaining partition of the aphasia dataset: 103 utterances that have been extracted from 6 transcripts these contain 37 word errors to be masked and repaired. These have been deliberately chosen to test a range of word types. For example, to observe how the model copes with the repair of words outside of its vocabulary, the test dataset includes masks that have been decomposed from 'complex' words. Such as the verb `[unfurls]` broken into `[un#]` `[#fur]` `[#ls]` and the contraction `[he's]` into `[he]` `[']` `[s]`. In order to make a prediction structured similar to the suggested ground truth, each repair is generated sequentially and evaluated independently. Due to the small size, 16% of the testing data contain these error cases. During each test the generated words are logged in a unique file and a master logfile is used to accumulate evaluation scores across all runs.

```
*INV: so — now I'm actually going to show you some pictures . and I'm
    gonna ask you to tell me a story &-um following them . and I need
    the story to have a beginning a middle and an end .
```

Transcript 5.1: A prompting utterance by the protocol's investigator (MMA002.cha).

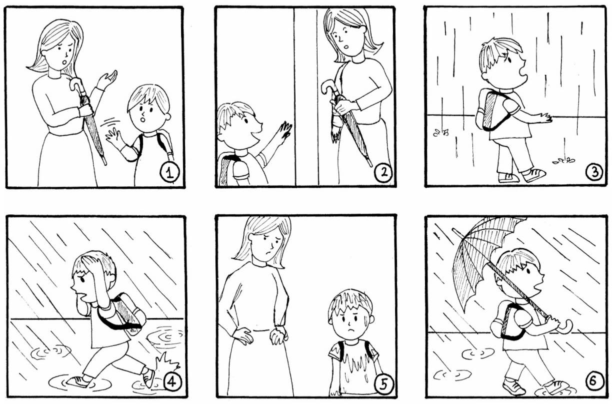


Figure 5.1: The input dataset consists of descriptive narrative transcriptions prompted by the image sequence 'Refused Umbrella' [38] [commissioned by AphasiaBank].

5.1 Key Findings

Weakness in the model has been observed when analysing language received by speakers with a high severity of aphasia, producing low confidence scores in the repair predictions. These speakers may produce heavily disjointed utterances containing a small number of words with minimal structure. These may rely highly on the use of visual material to support communication, for example the speaker points towards subjects in the image as a replacement for audible speech. Without the additional context of the images it is difficult to have high confidence when generating word replacements, as the model is unable to learn the inconsistent language patterns and is prone to instability.

With less severe cases of aphasia, the current models can produce word replacement repair can confidently be made to any substitution errors and areas of jargon that are related to the context of the trained narration task. For example, amending the noun substitution error `'snow' \(\rightarrow 'rain'\)`. Errors containing generalisable language and words out-side of BERT's vocabulary, require additional augmentation with context from neighbouring utterances to retrieve the correct repair.

Repair is successful on simple repetition errors in the cases where there is an exact pattern match of a word or phrase within the sequence: this can occur from hesitance and filled silences. Future exploration is needed into repairing repetition of non-perfect matched sequences. This would condense areas of perseverance where the speaker has worked to repair speech by removing sequential word types. For example, repetition caused by a stutter, `''he - he's - he is"` should be reduced to the single contraction to smooth the utterance further, as part of the inital cleaning.

All models have been observed to produce a consistent increase in accuracy when trained and tested on longer word sequences. When considering average performance the shortest input sequences - containing the pure extracted utterances - has produced an average accuracy of 36.9% across all models and baselines. Whereas, the use of augmentation to reach the largest sequence length requirement has increased this to an average of 55.2%.

5.2 Exploring Impact of the Training Domain

Table 5.1 presents the highest performances achieved for each model tested. This explores the observed impact when altering the domain context of the training dataset that has been used to fine-train the BERT model onto the current problem. The models used have taken a combination of transcription utterances extracted from the Williamson corpus [35] of speech produced by participants with aphasia, and the Capilouto corpus [36] containing control utterances of speakers without language or cognitive disorders. All utterances have been produced in response to the same narration prompt 5.1 as discussed in Section. 2.1.2.

The aggregated accuracies produced by each explored model are given in Figure 5.2 and 5.3 to visualise how model performance depends on the size of the context attached to the utterance. There is a necessity for additional training on the healthy contextual domain, with these models consistently producing an accuracy higher than calculated by the baseline models. The heat-maps illustrates change across the main evaluation scores: the rate of exact matches between the generated repair sequence and the human suggested ground truth, and the rate of responses that are considered valid synonyms.

The models that have not been further fine-tuned have achieved a best accuracy of 35.14% when performed on a full transcription gem, rather than individual utterances, showing that the short disjointed nature of aphasia speech present in the testing input data is not always sufficient to properly convey the speakers intent to the conversation partner. Increasing the size of the context used to train the model increases the accuracy of its predictions, utterances from the training dataset are padded to meet the length of the longest sequence.

The BERT models that have been fine-tuned purely on extracts of aphasia language data are able to greatly learn the language and speech patterns that are inherent in their communication, such as issues in sentence structure. These models have produced an accuracy of 62.16% when performed on input sequences of 20 words. Increasing the length of the aphasia data, when used as contextual information, does not always improve the performance of the model. Speech affected by aphasia typically contains heavily disjoint utterances; this may be due to severe topic changes and self-interruptions. At large concentrations, this may harm the loss function and cause overfitting of the model onto unrepresentative language patterns. The Williamson corpus contains a variation of aphasia quotients within the partitioned dataset. Therefore, participants within the training and testing sets may not present the same types of errors or struggle equally with narration speech. The impact different types of aphasia have on language production and comprehension throughout other language tasks should be explored in future adaptions of the project.

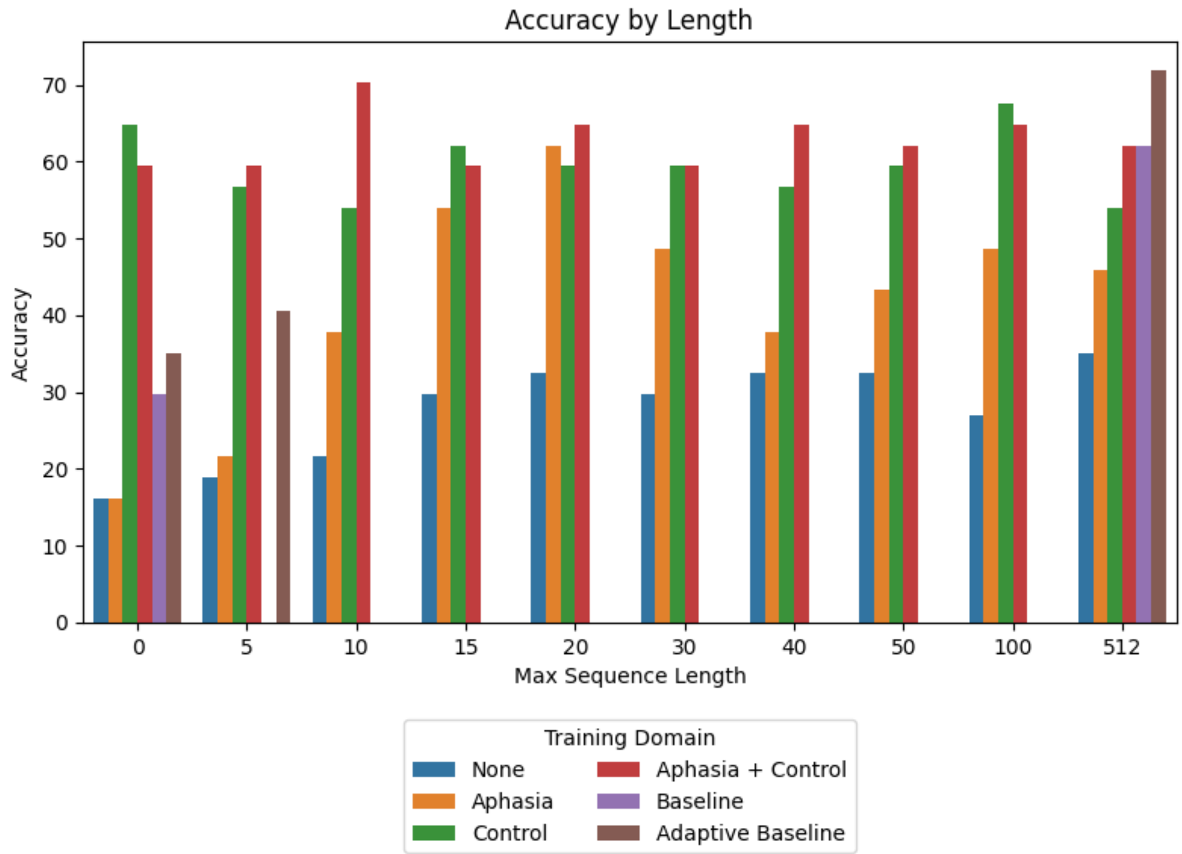


Figure 5.2: A visualisation of the performance of each model when using various sizes of data sequences during training and testing.

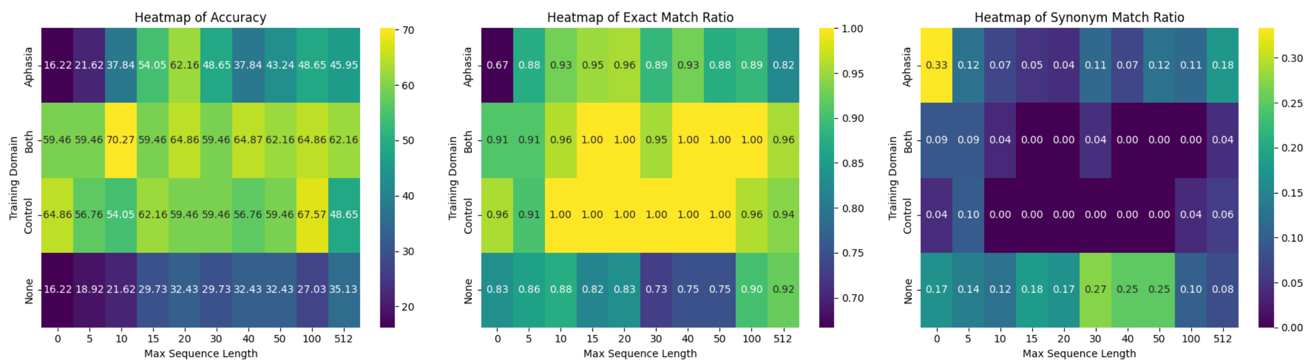


Figure 5.3: Heatmap overview of (a) the accuracy achieved from the model, (b) the ratio of exact matches and (c) synonym matches produced, using different fine-tuning approaches when executed on various sequence lengths.

The BERT models that have been fine-tuned using only the control utterances from the Capilouto dataset, has achieved a highest accuracy of 67.5%, this has been produced when executed on an input sequence length of 100 words. This suggests that the use of healthy control speech is capable of taking full advantage of the augmented data and the additional context that extends the utterances. This provides the model with useful information to learn expected language patterns at a vocabulary level: such as commonly occurring phrases when presented with the particular narration prompt. This will overfit to this testing exercise and will need additional training to generalise everyday speech.

The models that were trained on all available datasets achieved the best performance, an accuracy of 70.27%, on short utterances. In Figure 5.2 it can be seen that the model achieves similar performance scores on all input lengths. The model is able to take full advantage of both context types to perform without the need for large levels of data augmentation lengthening the input data. This could be able to produce translations within live conversations with little disruption [23], supporting the main objective of the project, encouraging conversation without introducing additional blocks.

Context Domain	Sequence Length	Accuracy (%)	Exact Ratio
None	512	35.14	0.92
Aphasia Speech	20	62.16	0.96
Control Speech	100	67.57	0.96
Control + Aphasia Speech	10	70.27	0.96
Baseline	512	62.1	1

Table 5.1: The key evaluation metrics for best performing models when fine-tuned on the available context domains.

5.2.1 Synonym Production

It has been observed that by increasing the size of context to inform the prediction there is a higher probability that the predicted word is an exact match to that that has been suggested by the human-transcriber - in the case where a successful repair has been made. In Figure 5.4 it can be seen that the proportion of exact matches within the valid cases follows an increasing trend as the sequence length increases. The models that have been trained on a combination of the control and aphasia data sets, have generated an exact match repair on 100% of their valid predictions, when performed on sequences longer than 10 words. Models trained on only the control dataset have similarly been found to produce the suggested repairs on sequences between 10 and 50 words. From these tests, this suggests that the addition of aphasia data into fine-tuning has assisted the prediction accuracy in cases where there is both too little and too much context available. Unlike the baseline tests, these models have been found to continuously produce the same predictions across multiple runs of the same input data.

A selection of the generated words for the best performing models are given within Table 5.2, during evaluation these are compared against the human suggested ground-truth, and the repair generated by the baseline models. For example, when provided with the pure utterance: `''a boy asks the woman she wants a [MASK]"`. The ChatGPT model that is used as a baseline has not received fine-tuning on the contextual domain, and therefore has no additional context to inform the prediction of this missing noun, and relies only on information within the augmented utterance. This is able to produce repairs such as `‘‘a boy asks the woman she wants a drink"` and `‘‘a boy asks the woman she wants a bag"`. The BERT models have otherwise been trained on narrations prompted by the same visual and audio prompts used to form the input data - given in Figure 5.1. Therefore, the model has an expectation of what nouns may be present within its vocabulary. This increases the model's produced confidence scores of the generated words and generates repairs with a higher contextual relevance, such as `‘‘a boy asks the woman she wants a umbrella"`. The baseline model is able to successfully predicted the noun replacement within augmented versions of the utterances as is able to use the additional context, on either sides of the masked word, as a reference for its predictions. For example, the following masked utterance `‘‘she puts the [MASK] into [MASK] hands "`, to produce `‘‘she puts the umbrella into his hands"`, as the noun 'umbrella' is repeated as a token within the surrounding transcription. All words predicted during the testing of the models can be found in the supplementary material.

Ground Truth	**Baseline Model**	**No fine-tuning**	**Aphasia Data**	**Control Data**	**Both Datasets**	**Success Rate**
ready	ready	ready	ready	ready	ready	100%
naughty	good	good	little	good	good	0%
umbrella	drink	date	umbrella	umbrella	umbrella	50%
he	he	he	he	he	he	92%
'	is	he	he	he	'	28.6%
s	running	him	s	starts	s	28.6%
umbrella	umbrella	umbrella	umbrella	umbrella	umbrella	85.7%
his	his	the	his	her	his	57.1%

Table 5.2: A comparison of sample words generated by the best performing models across each training domain: the ChatGPT baseline model; and the BERT models with no fine-tuning, fine-tuning on either aphasia or control transcripts, and fine-tuning on both sets of data. The percentage of tested BERT models that have generated a valid repair for this word: aggregated across the utterance level, paragraph level, and best performing sequence length. This illustrates how ease of prediction is dependent on word type as well as surrounding context.

5.3 Exploring Impact of Sequence Length

Additional model testing has been performed to explore the accuracy of predictions when processing larger segments of data, providing the MLM a higher level of context to inform word generation. This has been implemented following a similar approach to that of Binta et al. [15] who evaluated their models at both a utterance and paragraph level. The variation of sequence lengths have been achieved by placing a sliding window upon the datasets, this allows neighbouring segments to be used as padding on for the utterance to reach a minimum length boundary. The BERT model is bidirectional and can use the context on either side of the masked token to predict a suitable replacement word.

These tests consider the impact across all combinations of the explored training data domains, to observe performance across both healthy and aphasia effected speech. Speech produced by those with aphasia often contains very short disjointed utterances, models trained on this data alone has been found to produce low repair accuracies due to the lack of surrounding information to inform the prediction of replacement words. This has produced an accuracy of 16.2% compared to the 59.4% achieved when introducing healthy speech to the training set. The augmentation of longer sequences allows for greater confidence scores to be produced based on the larger surrounding context. The combination of aphasia and control datasets has been observed to produce the highest accuracies across each sequence length tested, until exceeding 100 words. At this point, the introduction of aphasia effected speech begins to harm the learning rate, the effect of this on larger speech segments must be considered in the next stage of investigations. This may be since bias is created towards language patterns specific to the individual speakers and the aphasia quotients involved. The disjoint nature of the speech also risks the introduction of irrelevant contextual information from neighbouring utterances that may harm the loss function of the model during the learning stages.

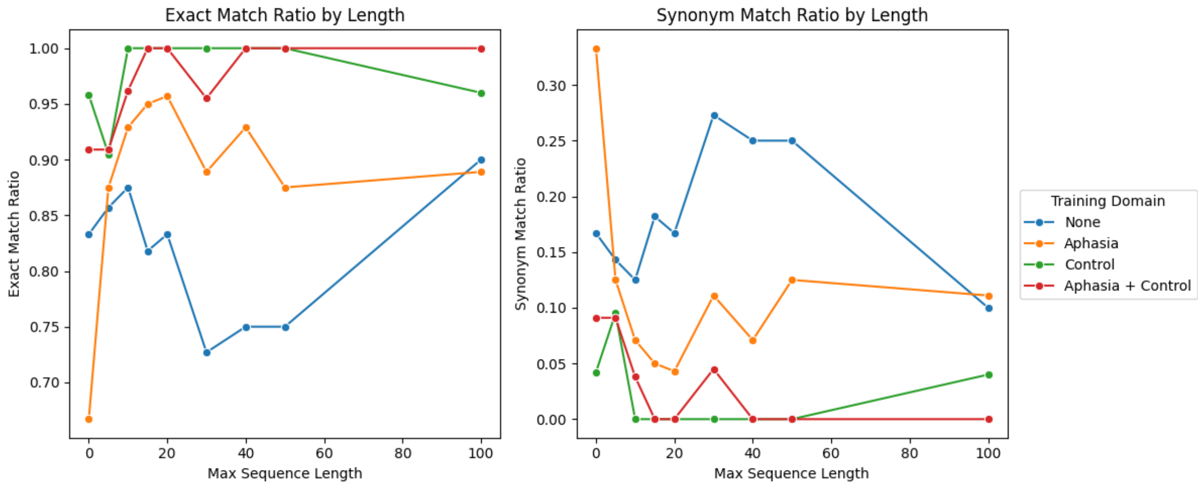


Figure 5.4: The evaluation of the generated predictions based upon the training domain, this is tested across different sequence lengths. Plot (a) visualises the Ratio of exact matches within the valid predictions. Plot (b) visualises the ratio of synonym matches. These are inverses by nature.

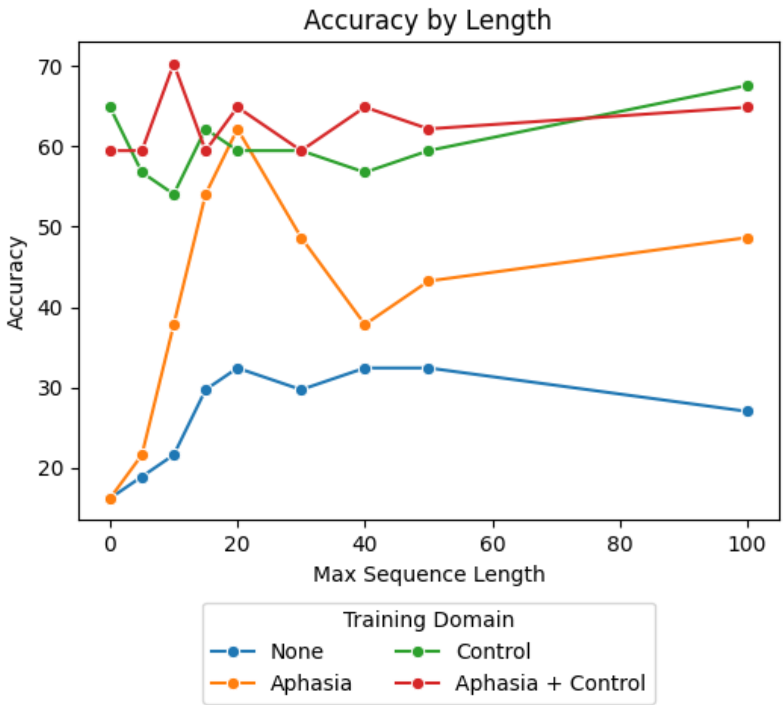


Figure 5.5: The accuracy of generated repair predictions across different sequence lengths used for training and testing data. Scores are aggregated for four of the BERT model types, line colour is used to differentiate these based upon their training datasets. This uses the Capilouto corpus [36] of healthy speakers as the control set, and the Williamson Corpus [35] aphasia speakers.

5.3.1 Testing at Utterance Level

During tests with minimum available context, the models have been trained on the pure utterances extracted from the transcripts, these may have a minimum length of only 1 word and a maximum of 30. The control datasets from the Capilouto corpus [36] have an average length of 6.7 words per utterance, while the aphasia utterances taken from the Williamson Corpus partitions [35] have an average of only 4.6 words used for training and 5.9 words for testing. The results produced from these tests are summarised in Table 5.3.

The HuggingFace pre-trained model used for the baseline value produced an accuracy of 16.22%, and fine-tuning on only the small partition of aphasia data makes no significant impact - though has increased the number of synonyms produced, as can be seen within Figure 5.4. This is much lower than the baseline performance when using minimum context of 29.7%, suggesting that, in the case where limited fine-tuning data is available, it is more reliable to use a commercial chatbot to repair the utterances.

Fine-tuning the model on a domain allows for further contextual information to be introduced without the need to augment the input data with longer utterances, or need additional preprocessing. This enables the specialisation of the model's vocabulary to weight confidence scores to suit the given situation. The use of the control Capilouto dataset has increased prediction accuracy to 64.86%, this is as the model has been allowed to specialise on the expected vocabulary using the example healthy passages. The increase in this accuracy can been seen in the prediction of nouns - for example, choosing `'umbrella'` rather than `'brush'`. Both nouns are chosen with high confidence by their models as create a valid sentence on substitution, but the use of the noun `'umbrella'` is more appropriate when given additional narrative context. This could be aided by the use of the visual prompt given in Figure 5.1, as discussed in Section 6.3.

Training Domain	**Min. Input Length (num. words)**	**Prediction Accuracy (%)**	**Ratio of Exact Matches**
None	0	16.22	0.83
Aphasia Speech	0	16.22	0.67
Control Speech	0	64.86	0.96
Control + Aphasia Speech	0	59.46	0.91
Baseline	0	29.73	1.00

Table 5.3: The model accuracies and their evaluation scores when executed on the extracted utterances without augmentation.

Target Word	**Single Utterance**	**Augmented Utterance**
ready	he's [MASK] to go .	and the boy is ready for school . he's [MASK] to go . and he says
naughty	you are a [MASK] boy .	and then his mother looks at him and says you are a [MASK] boy . you should have taken it .
umbrella	a boy asks the woman she wants a [MASK] .	a boy asks the woman she wants a [MASK] .
he's	so [MASK] [MASK] [MASK] running home in the rain .	so [MASK] [MASK] [MASK] running home in the rain .
umbrella, his	she puts the [MASK] into [MASK] hands .	the child comes home and the mother sees that the child is wet . she puts the [MASK] into [MASK] hands . and the child goes out and unfurls the umbrella .

Table 5.4: A selection of masked utterances and the ground truth suggested by the manual repair. These have been augmented to reach a minimum length of 10 words by concatenating neighbouring utterances by the same speaker within the transcript gem. Example predictions generated from these sequences can be found in Table 5.5.

Suggested Repair	Utterance Level	Moderate Context	Transcript Level
ready	ready	ready	ready
naughty	little	little	young
umbrella	umbrella	umbrella	drink
he ' s	he ' goes	he he s	he he is
umbrella his	umbrella his	umbrella his	umbrella the

Table 5.5: A sample of the words generated by the BERT model that has been trained on only repaired transcripts of aphasia narrations and their corresponding suggested ground truths. Output has been shown from three levels of context: the bare utterance, augmented context, and the full spoken gem. The highest accuracy has been achieved, from this model, when restricting utterances to a minimum size of 20 words.

5.3.2 Testing with Augmented Context

The length of utterances within the training and testing datasets have been explored for each of the contextual domains. This follows a similar technique used in the research of Binta et al., where utterances are restricted to only those greater than 5 words, to ensure enough meaningful context is introduced to the model analysis [15]. For this project, the length restriction is explored up to a maximum lower limit of 100 words, to observe BERT's context requirement for natural mask repairs. The final use-case focuses on live-speech production, therefore shorter paragraphs will reduce the impact on conversation flow since less data will be collected for significant output. More research is needed on larger areas of speech with finer granularity. The utterances have been augmented through the stitching of neighbouring segments of speech until the minimum word length requirement is reached. To reduce disruption, this padding only uses utterances that have been produced by the main speaker and are within the same transcription gem, and have the same topical context. A selection of utterances and example augmentations are provided in Table 5.4. As can be seen, this augmentation has avoided the dissection of speech segments, which may produce sequences of varying lengths depending on the size of the introduced lines.

The models that have completed their fine-tuning using both domain datasets have been found to produce the most accurate predictions when performed on shorter utterances, of minimum length 10 words: producing a success rate of 70.27%. The results of these tests are visualised in Figure 5.5, full results have been provided in the supplementary material. It has been observed that models trained using only the partitioned aphasia dataset have been found to be more successful on sequences that contain twice the amount of context, with a length of 20 words. This has given a score of 62.16%, almost twice the best success rate found without additional domain fine-tuning (32.43%).

Models that have been trained using only the control dataset have produced similar accuracies on shorter context sizes of 15 words. This is likely since a model trained on a smaller sample size, requires additional data augmentation to introduce a similar level of knowledge and achieve comparable results. It has been observed that increasing the length of the input sequences has had the greatest impact on models using aphasia transcripts for fine-tuning, a sample of the qualitative evaluation produced by this model can be seen in Table 5.4. Unlike models that depend on the healthy transcripts, the introduction of additional data does not always lead to an increase in word production accuracy. Instead, concatenating disjointed utterances from aphasia speech can lead to the learning of irrelevant language patterns due to the high error rates that may be present.

5.3.3 Testing at GEM Level

Tests on the full available context has been restricted to the maximum sequence length that the module can take as a single input, a length of 512. As some transcription gems are longer this, the have needed to be cropped to match the size of the input tensor. Augmentation has been applied by appending lines to either side of the utterance until the maximum length has been reached, to create a sliding window across the transcription. If the available transcription is shorter than this, the utterances are padded by the tokeniser using specialised token values.

The results produced from these tests are summarised within Table 5.6. When using no additional model fine-tuning, an accuracy of 35.14% is achieved, this is increased to 45.95% when aphasia data is introduced, and doubled to 62.16% when the control data is also used, giving an equal performance to the chatbot baseline. Therefore, a commercial chatbot would be the most appropriate tool in the case where the entire transcript of the speech is available for repair, as no additional data would be required to train the model to the same level. However, this is unlikely to be effective on live-speech repair due to the need to collect the entirety of the speech, rather than dynamically processing the utterances as they are produced; this would cause distribution to the fluidity of the conversation and may introduce additional barriers. Introducing additional context from the spoken prompt by the investigator has not increased performance of the model and has taken away useful context from the speaker, which is already limited. Future research could focus on the different contexts available to the model, such as simultaneous audio, visual, and text prompts to achieve dynamic results, inspired by the work of Ayesha et al [52]. This can explore the impact on repair accuracy and how this compares to the use of using context from additional utterances. This is discussed further in Section 6.3.

Training Domain	**Min. Input Length (num. words)**	**Prediction Accuracy (%)**	**Ratio of Exact Matches**
None	512	35.14	0.92
Aphasia Speech	512	45.95	0.82
Control Speech	512	54.05	1.00
Control + Aphasia Speech	512	62.16	0.96
Baseline	512	62.16	1.00

Table 5.6: The model accuracies and evaluation scores when the input sequence is augmented to meet the greatest minimum length restrictions.

Chapter 6

Conclusion

The adapted BERT models have been successful in predicting a stoppable substitution for a masked token based upon the surrounding words preceding and following it. This explores the bidirectional nature of the model's process by expanding the context available on either side of the masked token by stitching neighbouring utterances together. This increases prediction accuracy when applied on speech produced by speakers with no language impairments. Whereas, combining utterances that have been produced by a speaker with aphasia has been found to harm the model in some cases - due to a jump in conversation topics or highly unstructured phrases that may cause a fixation on ungeneralisable patterns.

This project explores the performances of the models when trained on varied contextual domains using data from the Aphasia Bank database. This includes the Williamson corpus of speech produced by participants with aphasia [35] and the Capilouto corpus containing speech from participants that have no language or cognition disorders [36]. These are captured following the same protocol and prompts, and are used as a control database to form a baseline to compare how BERT models perform on healthy expected speech. The BERT models that have received training on only the partitioned data from the aphasia speakers, have been found to produce its most accurate repairs when executed on shorter input sequences, performing best on utterances of 20 words with an accuracy of 62.16%. While the control training set has performed the best on sequences of 100 words with a similar accuracy of 67.57%. This is supported by the research of Binta et al [15], who have found that although training BERT models at a paragraph-level can increase performance on unseen scenarios, it can cause degradation if irrelevant information is included within context. They suggest initially training at a sentence level and introducing additional context to expand the models application on completion [15].

In general, it has been found that introducing the example transcripts into the training stage is important to generate reliable predictions. This produces a model that is well suited for repairing the responses to narration tasks, compared to when trained on only the aphasia speech. This may be due to the focus on word substitution repairs rather than structural repairs of the sentence and the importance of understanding the conversation topic. More reliable knowledge can be extracted from the healthy speech since the repaired aphasia transcripts are likely to contain irrelevant vocabulary from topic divergence.

6.1 Limitations of the Project

The difference that has been observed between the performance of the two training domains may be due to the size of the training data available for the model to learn from. The control training has taken place on 30 pieces of Capilouto data, while the aphasia set contains only 13 pieces of Williamson's repaired aphasia data after partitioning. The datasets containing data received by people with aphasia are often at a small scale due to the inherent challenges on collecting the data in a structured and repeatable way as must cater towards the difficulties common with language and comprehension disorders. To combat this, transcripts can be taken from additional AphasiaBank dataset's in combination to widen the fine-tuning and further test the impact of the training domain.

The size of the dataset limits the type of language errors that are represented within the chosen transcripts and restricts the exploration into how these errors are presented within speech. The Williamson dataset contains participants with a range of aphasia quotients, the type and severity of aphasia, these will have difficulties in different areas of language and production of different word types. Additional model training is necessary to cover other language tasks. Some transcripts within the Williamson data are much more disjoint with significantly shorter utterances, and others may have a larger number of word errors. The distribution of these severities across the training and testing partitions will have a large impact on the success of the model.

6.1.1 Words with Multiple Masks

During tokenisation of the utterance, words outside of the tokeniser's vocabulary are decomposed into multiple tokens, to allow inclusion of all words [1]. Alongside this, the suggested ground truths are tokenised to form a labelled validation tensor with a one-to-one mapping to the masked utterance. For example, the contraction 'he's' is mapped onto `[he]-[']-[s]`. The BERT model is able to predict one mask token at a time, and so may predict independent words where the suggested ground truth spans multiple mask tokens. The models can confidently predict single-token words, but are less accurate generating words outside of its vocabulary, likely to predict a synonymous sequence rather than an exact match. This is its main limitation when comparing against baseline models, such as ChatGPT, which are much more confident in this. The `SpanBERT` extension of BERT can instead be used to evaluate groups of tokens together to predict a cohesive solution.

6.1.2 Evaluating Synonyms

The evaluation function is unsuitable for words that have been decomposed into multiple mask tokens. For example, within the utterance `‘‘he unfurls the umbrella"` the verb is broken down into three masks `[un]-[\#\#ful][\#\#rs]`. The ChatGPT baseline has produced `‘‘he [opens]-[up]-[the] umbrella"`, when able to use maximum available context from all test runs. Even though the sentence reads with equivalent meaning, the one-to-one mask evaluation function would fail.

As a solution to this, paraphrase identification modules to analyse the implied definition of two sentences to detect equivalency [53]. This can be applied to compare the ground truth utterance against the generated one to perform the evaluation on a wider scale and consider the full sentence as a whole. This is especially useful for utterances that have multiple word errors that may propagate between disjointed utterances, and will help to ensure that information is not lost during speech repair. This should first be developed on control sequences and then specialise for the aphasia data, as this will inherently bring a lower confidence due the high error rate. Future evaluation of the final system is needed on live users, for this extra consideration must be made to use carefully crafted question prompts to ensure that both parties can be understood with ease [17].

6.2 Implications and Potential Applications

This project has been developed as exploration into an aid to smooth utterances produced by people with aphasia to facilitate conversation and regain confidence in their language abilities. Those with language impairments, such as aphasia, struggle with the comprehension and production of both spoken and written language. This can stem difficulties with frustration from themselves and their conversation partners, causing them to become socially isolated. It is suggested that therapeutical work focuses on finding comfort in their "new normal" and and return to their lives, routines, and relationships [9]. The use case of the project focuses on participants without co-existing cognitive disorders, such as dementia. These people often they have full awareness and have same knowledge and ideas they have always had, but now struggle to convey them - resulting in high likelihood for post-stroke depression [10].

This project has use as a communication aid and as a therapeutic tool, to provide a unique perspective into how speech is being interpreted by a conversation partner; giving insight into how successfully a speaker is able to convey their thoughts. This would normally require a trained individual to provide unfiltered feedback on their utterances, which would not always be available and adds an additional layer of interpretation. This provides optional monitoring of their recovery and therapy exercises that can be completed independently, outside of scheduled clinical hours. Tools such as this should never be used as a replacement to in-person therapy, due to the importance of face-to-face conversations on wellbeing, and instead be used as an additional tool alongside this.

Users would only require general technology training, due to the minimal necessary direct interaction with an interface, unlike using a chatbot such as ChatGPT or equivalent. Effort has been made to simplify necessary interactions due to the the known difficulties many experience when interacting with natural-language based interfaces. For example, research collected by Yoa et al. shows that those with language disorders face extensive discomfort during these interactions, finding difficulty in adapting their utterances to become recognisable and successfully navigate voice-assistants such as the Amazon Alexa [29]. Like many communication devices, successful use of the project relies on the available support from the conversation partner, as they will require patience to wait for the repaired output and their willingness to communicate alongside the tool [16].

6.3 Potential Future Work

The current stage of the project bases its evaluation upon the transcriptions to take advantage of the human interpretation of each utterance. The transcriber has included annotations to indicate areas of fault and differentiate the types of errors present. The speech within the Aphasia Bank corpus has been collected by TalkBank researchers with purpose to evaluate these errors across the aphasia demographics.

6.3.1 Locating Word Errors

Work exploring the classification of aphasic word production, such as that by Fraiser et al., use acoustic features that have been extracted from audio recordings to validate patterns against those produced by a healthy speaker [12]. This allows inference that the correct word has been chosen or pronounced in a recognisable way. A similar approach to data processing could be used to prepare audio data as an input to the project. Pattern analysis of these acoustic features would detect segments of speech that are likely to contain errors, such as areas of hesitation, stutter, or repetition. During the development stage this must consider intentional silences within the audio that have been created to de-identify the speech, such as removal of last names or addresses [40].

The final use case of this project would take input from live speech to adaptively assist real time conversations. There is ongoing research on the adaptation of ASRs to be suitable for disjointed speech produced by speakers with aphasia, due to the high use of jargon and mispronounced words [41] These ASRs often produce the transcribed speech formatted as XML, JSON, or TXT files, tokenising the utterance alongside its score of confidence recognition [41].

6.3.2 Non-Verbal Communication

The AphasiaBank collection protocol encourages their interviewers to use non-verbal prompts during data collection alongside scripted language - for example indicating towards the visual material and the use of gestures (such as head nods and eye contact) rather than verbal encouragers like "mhm" fillers [37]. This is stated as to be to avoid cluttering the manual transcription [37], and help to elicit successful utterances from the participants. This is likely to be as people with aphasia often make use of additional language techniques to communicate and aid comprehension of verbal prompts.

The use of non-verbal actions executed by the aphasic participant often provides essential support for their utterances, and may be necessary to fully smooth communications. An example of this is given with the transcript in Figure 6.1. The analysis of this is a current area of research by TalkBank [54]. Future expansion of this project could make use of this unspoken language to help communicate the user's intentions. During live speech, this would require a camera and computer vision techniques to analyse the user's actions [33]. For example, gestures and physical material are widely used to support speech, often used in attempt to communicate a word that the speaker is having difficulties producing.

```
*PAR: &=points:picture_three he's [/] &+tr &+k it's dry [: wet] [* s:
   r] and he [//] it [//] (..) it's raining [/] raining .

*PAR: &=points:picture_four look at her [: him] [* s:r:gc:pro] .

*PAR: &+i &+i (.) &-uh (.) &=points:picture_four then [/] &+h hen [:
   then] [* p:w] she [: he] [* s:r:gc:pro] was home .
```

Transcript 6.1: The speaker is using gestures and the supplied physical material to support their utterances and replace missing verbal language.

6.3.3 Increasing Context of Utterances

For some cases, a correct word has been attempted but is indistinguishable has been slurred or mispronounced - when repairing these, a phonetic transcription could be used to weight prediction and infer the intention with a higher confidence . This is available through an additional tier from CLAN transcripts in the AphasiaBank, and is often a necessity when manually transcribing jargon [40]. Another approach could make use the additional media that is provided alongside the written transcriptions in the AphasiaBank, as an additional input into the model.

Video Input

The AphasiaBank corpus includes video recordings of the interview, with focus on the aphasia speaker. This can be used to help visually identify of the areas of speech that are likely to have errors, such as word finding or production mistakes. This would help to automate the masking of erroneous words, rather than relying on the transcription annotations, and would help to expand the model onto live-speech production. Emotion recognition software, such as the `Aria` platform [55], can be used to monitor the movements within the speaker's facial expressions and body language to analyse for indications of uncertainty or frustration. These signs could indicate that the patient is struggling with their speech and will need assistance. Approaches that rely on live-video input would have a large limitation by the lack of availability of these media records, for training and development of the model, and the ethical considerations stemming from behaviour and emotion predictions.

Image Input

The use of images as an additional input to the model allows increases the context to increase the confidence of its predictions, as discussed within Section 3.4. Additional context could be taken from these video frames to infer context from computer vision analysis on the background setting and any actions or objects that can be recognised. Although this would be beneficial on live-use cases, the AphasiaBank data does lend itself towards this as the interviews are often taken place within a clinical setting, which does not correspond to the context of the utterances. Instead, the interviews make use of visual materials to prompt participants to narrate or recall their speech segments. Since all data collected through the corpus follows the same protocol, these images are available to be used as input into the model.

AI image generation tools, such as CLIP, can produce visual representation of the utterances and could be used to augment the dataset to introduce additional visual context that extrapolates from the disjointed transcriptions. Along with the translated coherent sentence, this could be provided to the conversation partner as a visual explanation of the speaker's intent, to help for understanding in the case that errors have not been completely removed.

6.4 Reflection

During development, the expected schedule has been closely followed and all key work packages produced. There has not been full implementation of the multimodal additions, though these have been explored.

The original project plan included automating the location of errors based on the generated word score under a threshold. Rather than using all available transcript annotations, only the word-substitution labels have been used to identify the error directly. This is because dynamic location is not necessary for this stage of the project and is more suitable when live-speech processing is implemented. This will be able to make use of the confidence labels that are assigned to the transcription of each word produced by the ASR. This can be applied in conjunction with located areas of repetition, hesitation, and self-interruption; that can be identified within the audio file, to highlight both transcription and speaker errors.

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